MIPS Reference Data ①

CORE INSTRUCTION SET

NAME	MNEMONIC	FORMAT	OPERATION (in Verilog)		OPCODE/ FUNCT (Hex)
Add	add	R	R[rd] = R[rs] + R[rt]	(1)	$0 / 20_{hex}$
Add Immediate	addi	I	R[rt] = R[rs] + SignExtImm	(1)(2)	8_{hex}
Add Imm. Unsigned	addiu	I	R[rt] = R[rs] + SignExtImm	(2)	9_{hex}
Add Unsigned	addu	R	R[rd] = R[rs] + R[rt]		$0 / 21_{hex}$
And	and	R	R[rd] = R[rs] & R[rt]		$0 / 24_{hex}$
And Immediate	andi	I	R[rt] = R[rs] & ZeroExtImm	(3)	c_{hex}
Branch On Equal	beq	I	if(R[rs]==R[rt]) PC=PC+4+BranchAddr*4	(4)	4_{hex}
Branch On Not Equal	bne	I	if(R[rs]!=R[rt]) PC=PC+4+BranchAddr*4	(4)	5_{hex}
Jump	j	J	PC=JumpAddr	(5)	2_{hex}
Jump And Link	jal	J	R[31]=PC+4;PC=JumpAddr	(5)	3_{hex}
Jump Register	jr	R	PC=R[rs]		$0 / 08_{hex}$
Load Byte Unsigned	lbu	I	R[rt]={24'b0,M[R[rs] +SignExtImm](7:0)}	(2)	24_{hex}
Load Halfword Unsigned	lhu	I	R[rt]={16'b0,M[R[rs] +SignExtImm](15:0)}	(2)	25_{hex}
Load Upper Imm.	lui	I	R[rt] = {imm, 16'b0}		f_{hex}
Load Word	lw	I	R[rt] = M[R[rs]+SignExtImm]	(2)	23_{hex}
Nor	nor	R	R[rd] = ~ (R[rs] | R[rt])		$0 / 27_{hex}$
Or	or	R	R[rd] = R[rs] | R[rt]		$0 / 25_{hex}$
Or Immediate	ori	I	R[rt] = R[rs] | ZeroExtImm	(3)	d_{hex}
Set Less Than	slt	R	R[rd] = (R[rs] < R[rt]) ? 1 : 0		$0 / 2a_{hex}$
Set Less Than Imm.	slti	I	R[rt] = (R[rs] < SignExtImm) ? 1 : 0	(2)	a_{hex}
Set Less Than Imm. Unsigned	sltiu	I	R[rt] = (R[rs] < SignExtImm) ? 1 : 0	(2)(6)	b_{hex}
Set Less Than Unsigned	sltu	R	R[rd] = (R[rs] < R[rt]) ? 1 : 0	(6)	$0 / 2b_{hex}$
Shift Left Logical	sll	R	R[rd] = R[rt] << shamt		$0 / 00_{hex}$
Shift Right Logical	srl	R	R[rd] = R[rt] >> shamt		$0 / 02_{hex}$
Store Byte	sb	I	M[R[rs]+SignExtImm](7:0) = R[rt](7:0)	(2)	28_{hex}
Store Halfword	sh	I	M[R[rs]+SignExtImm](15:0) = R[rt](15:0)	(2)	29_{hex}
Store Word	sw	I	M[R[rs]+SignExtImm] = R[rt]	(2)	$2b_{hex}$
Subtract	sub	R	R[rd] = R[rs] - R[rt]	(1)	$0 / 22_{hex}$
Subtract Unsigned	subu	R	R[rd] = R[rs] - R[rt]		$0 / 23_{hex}$

(1) May cause overflow exception
(2) SignExtImm = { 16{immediate[15]}, immediate }
(3) ZeroExtImm = { 16{1b'0}, immediate }
(4) BranchAddr = { 14{immediate[15]}, immediate, 2'b0 }
(5) JumpAddr = { PC[31:28], address, 2'b0 }
(6) Operands considered unsigned numbers (vs. 2 s comp.)

BASIC INSTRUCTION FORMATS

R	opcode	rs	rt	rd	shamt	funct
	31–26	25–21	20–16	15–11	10–6	5–0
I	opcode	rs	rt	immediate		
	31–26	25–21	20–16	15–0		
J	opcode	address				
	31–26	25–0				

REGISTER NAME, NUMBER, USE, CALL CONVENTION

NAME	NUMBER	USE	PRESERVED ACROSS A CALL?
$zero	0	The Constant Value 0	N.A.
$at	1	Assembler Temporary	No
$v0-$v1	2-3	Values for Function Results and Expression Evaluation	No
$a0-$a3	4-7	Arguments	No
$t0-$t7	8-15	Temporaries	No
$s0-$s7	16-23	Saved Temporaries	Yes
$t8-$t9	24-25	Temporaries	No
$k0-$k1	26-27	Reserved for OS Kernel	No
$gp	28	Global Pointer	Yes
$sp	29	Stack Pointer	Yes
$fp	30	Frame Pointer	Yes
$ra	31	Return Address	No

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